

EXPERIMENT 5

Aim:

Create SQL views for department salary summary and employee hierarchy. Test updatability of views. Implement a recursive CTE to display reporting chains.

Source Code:

```
CREATE TABLE Department (  
    dept_id INT PRIMARY KEY,  
    dept_name VARCHAR(50),  
    location VARCHAR(50)  
);  
  
CREATE TABLE Project (  
    project_id INT PRIMARY KEY,  
    project_name VARCHAR(100),  
    budget DECIMAL(12,2)  
);  
  
CREATE TABLE Employee (  
    emp_id INT PRIMARY KEY,  
    emp_name VARCHAR(50),  
    job_title VARCHAR(50),  
    salary DECIMAL(10,2),  
    hire_date DATE,  
    dept_id INT,  
    manager_id INT,  
    FOREIGN KEY (dept_id) REFERENCES Department(dept_id),  
    FOREIGN KEY (manager_id) REFERENCES Employee(emp_id)  
);  
  
CREATE TABLE Employee_Project (  
    emp_id INT,  
    project_id INT,  
    hours_worked INT,  
    PRIMARY KEY (emp_id, project_id),  
    FOREIGN KEY (emp_id) REFERENCES Employee(emp_id),
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```

FOREIGN KEY (project_id) REFERENCES Project(project_id)
);

INSERT INTO Department VALUES
(1, 'IT', 'Delhi'),
(2, 'HR', 'Gurgaon'),
(3, 'Finance', 'Mumbai'),
(4, 'Marketing', 'Bangalore'),
(5, 'Sales', 'Hyderabad');

INSERT INTO Project VALUES
(101, 'Cloud Migration', 500000),
(102, 'E-Commerce Website', 350000),
(103, 'Mobile Application', 250000),
(104, 'HR Automation', 150000),
(105, 'Financial Analytics', 450000),
(106, 'Marketing Campaign', 200000),
(107, 'CRM Development', 300000),
(108, 'Cyber Security', 600000);

INSERT INTO Employee VALUES
(1, 'Amit', 'IT Manager', 90000, '2018-01-10', 1, NULL),
(2, 'Neha', 'HR Manager', 85000, '2018-03-15', 2, NULL),
(3, 'Rahul', 'Finance Manager', 95000, '2017-06-20', 3, NULL),
(4, 'Priya', 'Marketing Manager', 88000, '2019-02-12', 4, NULL),
(5, 'Arjun', 'Sales Manager', 92000, '2017-09-25', 5, NULL);

INSERT INTO Employee VALUES
(6, 'Ravi', 'Software Engineer', 60000, '2020-01-10', 1, 1),
(7, 'Simran', 'Software Engineer', 65000, '2021-04-15', 1, 1),
(8, 'Karan', 'Database Developer', 70000, '2020-07-20', 1, 1),
(9, 'Pooja', 'Cloud Engineer', 75000, '2022-01-12', 1, 1),
(10, 'Vikas', 'System Engineer', 58000, '2021-09-25', 1, 1),
(11, 'Anjali', 'HR Executive', 55000, '2020-02-10', 2, 2),
(12, 'Rohit', 'Recruiter', 50000, '2021-05-15', 2, 2),
(13, 'Kavya', 'HR Executive', 52000, '2022-06-20', 2, 2),

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(14, 'Manish', 'Training Officer', 60000, '2020-08-12', 2, 2),
(15, 'Sneha', 'HR Analyst', 58000, '2021-10-25', 2, 2),
(16, 'Deepak', 'Financial Analyst', 70000, '2019-02-10', 3, 3),
(17, 'Nisha', 'Accountant', 62000, '2020-05-15', 3, 3),
(18, 'Varun', 'Financial Analyst', 68000, '2021-06-20', 3, 3),
(19, 'Meena', 'Accountant', 60000, '2022-08-12', 3, 3),
(20, 'Sahil', 'Finance Executive', 57000, '2021-11-25', 3, 3),
(21, 'Tanya', 'Marketing Executive', 55000, '2020-03-10', 4, 4),
(22, 'Mohit', 'SEO Specialist', 60000, '2021-05-15', 4, 4),
(23, 'Isha', 'Content Manager', 65000, '2022-07-20', 4, 4),
(24, 'Akash', 'Marketing Analyst', 58000, '2020-09-12', 4, 4),
(25, 'Riya', 'Social Media Manager', 62000, '2021-12-25', 4, 4),
(26, 'Naveen', 'Sales Executive', 55000, '2020-04-10', 5, 5),
(27, 'Komal', 'Sales Executive', 57000, '2021-06-15', 5, 5),
(28, 'Yash', 'Sales Analyst', 60000, '2022-07-20', 5, 5),
(29, 'Pankaj', 'Business Executive', 65000, '2020-10-12', 5, 5),
(30, 'Divya', 'Sales Executive', 58000, '2022-01-25', 5, 5);

INSERT INTO Employee_Project VALUES

(1,101,100),
(6,101,120),
(7,102,100),
(8,101,90),
(9,108,130),
(10,103,80),
(2,104,100),
(11,104,90),
(12,104,80),
(13,104,70),
(14,104,100),
(15,104,75),
(3,105,100),

(16,105,120),
(17,105,100),
(18,105,90),
(19,105,80),
(20,105,70),
(4,106,100),
(21,106,120),
(22,106,100),
(23,106,90),
(24,106,80),
(25,107,100),
(5,107,100),
(26,107,110),
(27,107,90),
(28,107,80),
(29,107,100),
(30,107,90);

CREATE VIEW department_salary_summary AS

SELECT

d.dept_id,

d.dept_name,

COUNT(e.emp_id) AS employee_count,

AVG(e.salary) AS average_salary,

SUM(e.salary) AS total_salary,

MAX(e.salary) AS maximum_salary,

MIN(e.salary) AS minimum_salary

FROM Department d

LEFT JOIN Employee e

ON d.dept_id = e.dept_id

GROUP BY d.dept_id, d.dept_name;

SELECT *

```

FROM department_salary_summary;

CREATE VIEW employee_hierarchy AS

SELECT

    e.emp_id,

    e.emp_name AS employee_name,

    e.job_title,

    m.emp_name AS manager_name

FROM Employee e

LEFT JOIN Employee m

ON e.manager_id = m.emp_id;

UPDATE Employee

SET emp_name = 'Ravi Kumar'

WHERE emp_id = 6;

SELECT *

FROM employee_hierarchy;

-- Recursive CTE

WITH RECURSIVE employee_chain AS

(

    SELECT

        emp_id,

        emp_name,

        manager_id,

        0 AS level,

        CAST(emp_name AS CHAR(500)) AS reporting_chain

    FROM Employee

    WHERE manager_id IS NULL

    UNION ALL

    SELECT

        e.emp_id,

        e.emp_name,

        e.manager_id,

        ec.level + 1,

```

```
        CONCAT(ec.reporting_chain, ' -> ', e.emp_name)
FROM Employee e
JOIN employee_chain ec
ON e.manager_id = ec.emp_id
)
SELECT
    emp_id,
    emp_name,
    level,
    reporting_chain
FROM employee_chain
ORDER BY reporting_chain;
```

Output:

Ctrl + K

I/O

AI Agent

99 ms

STDBIN

Input for the program (Optional)

Output

dept_id	dept_name	employee_count	average_salary	total_salary	maximum_salary	minimum_salary
1	IT	6	69666.66667	418000.00	90000.00	58000.00
2	HR	6	60000.00000	360000.00	85000.00	50000.00
3	Finance	6	68666.66667	412000.00	95000.00	57000.00
4	Marketing	6	64666.66667	388000.00	82000.00	55000.00
5	Sales	6	64500.00000	387000.00	92000.00	55000.00

emp_id	employee_name	job_title	manager_name
1	Amit	IT Manager	NULL
2	Neha	HR Manager	NULL
3	Rahul	Finance Manager	NULL
4	Priya	Marketing Manager	NULL
5	Arjun	Sales Manager	NULL
6	Ravi Kumar	Software Engineer	Amit
7	Simran	Software Engineer	Amit
8	Karan	Database Developer	Amit
9	Pooja	Cloud Engineer	Amit
10	Vikas	System Engineer	Amit
11	Anjali	HR Executive	Neha
12	Rohit	Recruiter	Neha
13	Kavya	HR Executive	Neha
14	Manish	Training Officer	Neha
15	Sneha	HR Analyst	Neha
16	Deepak	Financial Analyst	Rahul
17	Nisha	Accountant	Rahul
18	Varun	Financial Analyst	Rahul
19	Meena	Accountant	Rahul

Ctrl + K

I/O

AI Agent

99 ms

STDBIN

Input for the program (Optional)

Output

18	Varun	Financial Analyst	Rahul
19	Meena	Accountant	Rahul
20	Sahil	Finance Executive	Rahul
21	Tanya	Marketing Executive	Priya
22	Mohit	SEO Specialist	Priya
23	Isha	Content Manager	Priya
24	Akash	Marketing Analyst	Priya
25	Riya	Social Media Manager	Priya
26	Naveen	Sales Executive	Arjun
27	Komal	Sales Executive	Arjun
28	Yash	Sales Analyst	Arjun
29	Pankaj	Business Executive	Arjun
30	Divya	Sales Executive	Arjun

emp_id	emp_name	level	reporting_chain
1	Amit	0	Amit
8	Karan	1	Amit -> Karan
9	Pooja	1	Amit -> Pooja
6	Ravi Kumar	1	Amit -> Ravi Kumar
7	Simran	1	Amit -> Simran
10	Vikas	1	Amit -> Vikas
5	Arjun	0	Arjun
30	Divya	1	Arjun -> Divya
27	Komal	1	Arjun -> Komal
26	Naveen	1	Arjun -> Naveen
29	Pankaj	1	Arjun -> Pankaj
28	Yash	1	Arjun -> Yash
2	Neha	0	Neha
11	Anjali	1	Neha -> Anjali
13	Kavya	1	Neha -> Kavya

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NewHistorySave

Ctrl + K

I/O

AI Agent

99 ms

STDBIN

Input for the program (Optional)

Output

1	Amit	0	Amit
8	Karan	1	Amit -> Karan
9	Pooja	1	Amit -> Pooja
6	Ravi Kumar	1	Amit -> Ravi Kumar
7	Simran	1	Amit -> Simran
10	Vikas	1	Amit -> Vikas
5	Arjun	0	Arjun
30	Divya	1	Arjun -> Divya
27	Komal	1	Arjun -> Komal
26	Naveen	1	Arjun -> Naveen
29	Pankaj	1	Arjun -> Pankaj
28	Yash	1	Arjun -> Yash
2	Neha	0	Neha
11	Anjali	1	Neha -> Anjali
13	Kavya	1	Neha -> Kavya
14	Manish	1	Neha -> Manish
12	Rohit	1	Neha -> Rohit
15	Sneha	1	Neha -> Sneha
4	Priya	0	Priya
24	Akash	1	Priya -> Akash
23	Isha	1	Priya -> Isha
22	Mohit	1	Priya -> Mohit
25	Riya	1	Priya -> Riya
21	Tanya	1	Priya -> Tanya
3	Rahul	0	Rahul
16	Deepak	1	Rahul -> Deepak
19	Meena	1	Rahul -> Meena
17	Nisha	1	Rahul -> Nisha
20	Sahil	1	Rahul -> Sahil
18	Varun	1	Rahul -> Varun

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